Selected O-R: causal renewal and a complete analytical cohort

15 October 2026 — mathematical qualification; no traffic experiment

Status. The user selects renewal; RE-O1 closes. Qualify the previously specified recent-mass O-R formula, retaining R-E and S-E-P. No stateless replacement, step-dependent reset or new physical formula is introduced. The bounded recovery candidate/analytical pass is complete at **OR-Q1**. Whole-range event qualification, applied MP6 and Step 3 remain open. Parameters and representative speeds remain hypotheses.

1. The conditional member being qualified

Keep H1 preparation, cutoff, initially empty T1 and constant advice $v \in [v_-, v_+]$. For $j = A, E, B$,

$$P_j(t) = F_j(t - b/v), \quad q_j = P_j - Y_j, \quad N_j = F_j - Y_j, \quad \lambda_j = \dot{P}_j, \quad g_j = \dot{Y}_j. \tag{1}$$

The B lane is independent and unqueued. $B(t)$ denotes instead the shared A/E budget. With $K = wk_j$,

$$c(u) = \frac{Ku}{w+u}, \quad C(v) = c(v), \quad U(f) = \frac{wf}{K-f}, \quad 0 < R \le C(v_-), \quad u_R = U(R). \tag{2}$$

Use the existing trial work family and sufficient monotonicity domain:

$$c_A = 1, \quad c_E = c_0 + \beta \frac{g_A}{C(v)} \frac{[u_A - u_E]_+}{v}, \quad c_0 = 1 + \delta, \quad \eta = c_0 - \beta(1 + v_+/w) > 0. \tag{3}$$

Here $\delta, \beta \ge 0$, with free marks $u_j = v$ and queued marks $u_j = U(g_j)$. S-E-P gives a fitting free A offer priority; queued A uses the earlier equal-work/unused-work template. Its selected-regime inverse is unique under (3). These remain trial constitutive assumptions, not a field-calibrated fairness or crossing-speed law.

For positive physical acceleration a_r,

$$\begin{aligned} V(\theta; u) &= \sqrt{(w+u)^2 + 2wa_r\theta} - w, \\ D(\theta; u) &= [R - c(V(\theta; u))]_+, \quad \theta \ge 0, \\ W(u) &= \frac{k_c(u_R)[u_R - u]_+^2}{2a_r}, \quad k_c(z) = \frac{K}{w+z}. \end{aligned} \tag{4}$$

D decreases with age and mark, $0 \le D \le R$, $|D_\theta| \le a_r k_j$, $|D_u| \le k_j$, and expires within $\Theta_{\max} = u_R/a_r + u_R^2/(2a_r w)$. Write $W_0 = W(0)$. The age includes backward propagation and is independent of numerical steps.

2. The selected renewal formula and its physical-time update

Let $Y = Y_E$, $g = g_E$, and $m = m_{\text{ref}} = 1$ vehicle-equivalent. For the mass-rank inverse $T(\xi)$ of Y, define $a(t) = [Y(t) - m]_+$ and

$$p(t) = \frac{1}{m} \int_{a(t)}^{Y(t)} D(t - T(\xi); u_E(T(\xi)))d\xi, \qquad B(t) = [R - p(t)]_+. \tag{5}$$

This is the previously proposed O-R law. The reference mass is physical, not a numerical bin or integer admission. Only E creates profiles; A still affects E admission and current sharing. Traffic mass is never removed when a profile retires.

The equivalent time-history form is

$$p(t) = \frac{1}{m} \int_0^t D(t - s; u_E(s))g(s)\mathbf{1}_{\{Y(s)>a(t)\}}ds. \tag{6}$$

Flat portions of Y have zero admission measure, so inverse choices and rank endpoints do not change the integral. The history can be restricted to both age $\Theta_{\max}$ and the latest admitted mass m.

Where derivatives exist, let $T_o = T(Y(t) - m)$ when $Y(t) > m$. Then

$$\begin{aligned} \dot{p} = &\frac{g(t)}{m} D(0; u_E(t)) \\ &+ \frac{1}{m} \int_{a(t)}^{Y(t)} D_\theta(t - T(\xi); u_E(T(\xi)))d\xi \\ &- \frac{g(t)}{m} \mathbf{1}_{\{Y(t)>m\}} D(t - T_o; u_E(T_o)). \end{aligned} \tag{7}$$

The three terms are new entry, aging of retained profiles, and retirement of oldest weight. No division by the oldest discharge rate is needed. At $Y = m$, use the a.e. version; the cumulative account is continuous. With $g = 0$, only aging remains, so recovery continues without new discharge.

The reading alone generally does not determine future aging or retirement: recent age/mark information is part of the state. New fast E has zero new loss but can retire an old slow profile. This is the accepted renewal approximation to qualify physically, not an exact moving-leader solver. It does not reset at queue clearance or each numerical step.

3. Causal construction including the moving mass cutoff

Compare two admitted histories (g_i, u_i, Y_i) with identical past through t_0, $0 \le g_i \le G := R/c_0$, and $0 \le u_i \le v_+$. For fixed $t > t_0$, put

$$\epsilon = \sup_{s \le t} |Y_1(s) - Y_2(s)| \le \int_{t_0}^{t} |g_1 - g_2| ds, \qquad |a_1(t) - a_2(t)| \le \epsilon. \tag{8}$$

Use $f_i(s) = D(t - s; u_i(s))$ and the indicator I_i from (6). The difference with I_1 held fixed is bounded by $R \int |\Delta g| + k_j G \int |\Delta u|$. Where $I_1 \ne I_2$, $|Y_2(s) - a_2(t)| \le 2\epsilon$. The $g_2 ds$ mass in that rank band is at most 4ϵ, including plateaus. Thus

$$|p_1(t) - p_2(t)| \le \frac{1}{m} \left[5R \int_{t_0}^{t} |\Delta g| ds + k_j G \int_{t_0}^{t} |\Delta u| ds \right]. \tag{9}$$

This estimate needs no positive lower admission rate or smooth inverse. It also allows bounded discontinuities in already recorded speed marks.

On a valid selected node-regime chart, (3) gives local uniform Lipschitz constants L_g, L_u for g_E and its mark as functions of B with the same arrival data. Since the budget map is 1-Lipschitz,

$$\|\Delta \mathcal{B}\|_{\infty,[t_0,t_0+\tau]} \le \frac{\tau}{m} (5RL_g + k_j G L_u) \|\Delta B\|_{\infty,[t_0,t_0+\tau]}. \tag{10}$$

For sufficiently short τ, this is a contraction on a small valid chart neighborhood. Bounded account increments preserve that neighborhood until its next guard. Successive local intervals give causal continuation while the chart stays valid, and the same estimates give finite-interval continuous dependence there. The interval length is a proof device.

Equation (10) is not a proof of every endogenous queue switch or a claim that a free chart is valid when its offer fails. No fallback, numerical hysteresis or unapproved mode extension is used. The strict transition checks already established for S-E-P remain separate inputs. The finite event family below closes its own particular switching sequence.

4. Derived uniform positive service and finite clearance

The retained E mass is at most m and its rate at most $G = R/c_0$. Because $D(\theta; u) \le D(\theta; 0)$ decreases with age, its largest possible integral fills the most recent ages at rate G. Therefore

$$p(t) \le p_{\max} := \frac{G}{m} \int_0^{m/G} D(\theta; 0) d\theta < R, \qquad B(t) \ge B_{\min} := R - p_{\max} > 0. \tag{11}$$

The strict inequality follows from $a_r > 0$: $D(\theta; 0) < R$ at every positive age. The rearrangement bound is valid even when less than m has passed. It is uniform in time and advice for the fixed parameters. It is derived, not an added service floor; it can approach zero as recovery becomes arbitrarily slow. The auxiliary $a_r = 0$ limit below is excluded from this bound and the admitted model.

Let $\overline{M}_E = P_E(L/v)$, $Q = q_A + q_E$ and $c_{\max} = c_0 + \beta$. The O-R integrand is a subset of the additive one, so Tonelli gives

$$\int_0^\infty p(t) dt \le \overline{M}_E W_0/m. \tag{12}$$

For any admissible joined solution, work conservation after $t_* = L/v$ and (11)–(12) give either sufficient bound

$$\begin{aligned} T_{\text{clear}} &\le t_* + c_{\max} Q(t_*)/B_{\min}, \\ T_{\text{clear}} &\le t_* + \{c_{\max} Q(t_*) + \overline{M}_E W_0/m\}/R. \end{aligned} \tag{13}$$

These use available finite E mass before clearance, not assumed final completion. They do not prove existence of every joined solution.

Limits and simpler rivals. With no E, $p = 0$. When free offers fit from the H1 front, decreasing E offer keeps $p = q_A = q_E = 0$ under the causal free-mode guard. For a constant free A offer $a < R$, (7) gives $\dot{p} \le R g_E/m$ and $g_E \le (B - a)/c_0$. The gap $x = B - a$ therefore satisfies $\dot{x} \ge -Rx/(mc_0)$ and remains positive at finite times. Renewal preserves the constant-A protection limit; increasing A arrivals need a separate argument. For constant queued E rate and mark over the renewal window,

$$p = \frac{g_E}{m} W_{m/g_E}(u_E), \quad W_h(u) := \int_0^h D(\theta; u) d\theta, \quad g_A + c_E g_E = R - \frac{g_E}{m} W_{m/g_E}(u_E). \tag{14}$$

This is the restricted stationary target for a simpler state-dependent instantaneous rival, not its complete transient selection theorem. O-S, fixed M1, aggregate/history/scalar and search rivals remain. History must earn a consequential transient role; no interior optimum is required.

5. A nonempty H1 build-up with A delay: auxiliary prefix

This is an existence construction, not a chosen traffic scenario, calibration or optimization result. Fix one advice v_0 and the translated eligible pulse, initially with empty queues/account. Choose a donor-limited H1 front, so during the pulse

$$e(t) = e_f e^{-kt}, \quad a(t) = s - e(t), \quad k = v_0/h > 0, \quad s = v_0 r < R, \quad \varepsilon = R - s > 0. \tag{15}$$

Here a, e are *arrival rates*; h is the retained H1 opportunity length. Choose $c_0 > 1$, $e_f > 0$, $a_f = s - e_f > 0$ such that

$$\varepsilon < \delta e_f/2, \quad G_0 = (\varepsilon + e_f)/c_0 < R, \quad d_0 = R - G_0 > \varepsilon + e_f, \quad \delta = c_0 - 1. \tag{16}$$

Initially $a_f + c_0 e_f = R - \varepsilon + \delta e_f > R$, so the strict S-E-P front activates E while A stays free.

First set $a_r = \beta = 0$ solely as a finite-prefix auxiliary limit. Before the renewal window fills and while A is free and E queued, (5) becomes

$$g_E = \frac{B - a(t)}{c_0}, \quad \dot{B} = -\frac{g_E(R - g_E)}{m}, \quad \dot{q}_E = e - g_E. \tag{17}$$

The queued trace gives $D(0; U(g_E)) = R - g_E$. While A is free, $g_E \le G_0$ and $B \ge a \ge a_f$. Hence

$$R - B \ge d_0 Y_E/m, \quad Y_E \le m\frac{\varepsilon + e_f}{d_0} < m. \tag{18}$$

This rules out a first window fill before A onset, validating the prefix.

Until B reaches s, if $e \ge e_f/2$, then

$$-\dot{B} \ge \mu := \frac{e_f d_0}{2mc_0} > 0, \quad \dot{q}_E \ge (\delta e_f/2 - \varepsilon)/c_0 > 0. \tag{19}$$

Choose k sufficiently small that $k\varepsilon/\mu < \log 2$. Then B reaches s by ε/μ while that bound holds and passes strictly below s. Thereafter, while A remains free, $g_E < e/c_0 < e$, so E cannot clear. Since B is nonincreasing and $a(t) \uparrow s$, a finite first A onset t_A must occur. At that hit,

$$x = B - a = 0, \quad \dot{x} = -ke(t_A) < 0, \qquad \dot{q}_A(t_A+) = a(t_A) - B(t_A)/2 = B(t_A)/2 > 0. \tag{20}$$

The first equality uses the pre-onset side; the last uses queued cooperation.

6. Ensuring an E backlog and a nontrivial renewal event

The construction must exercise renewal, not only its additive initial prefix. At $k = 0$ in the same auxiliary input, A is constant a_f and the usual barrier keeps $B > a_f$ at every finite time. E stays queued with

$$\dot{q}_E \geq \sigma := (\delta e_f - \varepsilon)/c_0 > 0, \qquad Y_E < m. \tag{21}$$

Choose finite T_b with $\sigma T_b > 3m$. On this fixed prefix the free-A gap is strictly positive. Continuous dependence of (17) on the input makes it possible to choose a sufficiently small *positive* k so A stays free through T_b and $q_E(T_b) > 2m$. Condition (19) can be retained. The finite A onset from (20) is later, with still larger E backlog. No constant-input or $k = 0$ primary task is adopted.

End the actual finite pulse at $T > t_A$ sufficiently close to t_A. Then $q_A(T) > 0$, $q_E(T) > 2m$, $Y_E(T) < m$ and $B(T) > 0$. Choose T closer if necessary so

$$\frac{q_A(T)}{c_0} < m - Y_E(T), \quad q_A(T) < c_0 q_E(T), \quad \frac{R q_A(T)}{m c_0} < B(T)/2. \tag{22}$$

These inequalities are possible because $q_A(T) \downarrow 0$ as $T \downarrow t_A$ while the other margins remain positive.

In the auxiliary zero-recovery continuation, while both queues drain and the window is unfilled, $g_A = B/2$, $g_E = B/(2c_0)$. Thus A clears first, using at most $q_A(T)/c_0$ E mass, before window fill. Since $dB/dY_E = -(R - g_E)/m \geq -R/m$, (22) keeps $B \geq B(T)/2$ until that strict A clearance. Afterwards A has no arrivals and $g_E = B/c_0$.

The first $Y_E = m$ is reached at finite time in this auxiliary E-only prefix. Before it, $B \geq R(1 - Y_E/m) > 0$. If Y_E approached m only asymptotically, the additional identity

$$B = R(1 - Y_E/m) + \frac{1}{m}\int_0^{Y_E} g_E(T(\xi))d\xi \tag{23}$$

would give a strictly positive limiting budget, since some earlier E has passed at positive rate. That contradicts asymptotic approach. Thus fill occurs with $g_E > 0$ and a positive E backlog. The oldest profile then has positive loss $R - g_E(0+) > 0$. Immediately after fill, the retirement term in (7) is genuinely nonzero. This finite auxiliary prefix is all that is needed; its nonrecovering infinite tail is not an admitted model.

7. Positive recovery, positive mismatch and complete clearance

On every fixed finite interval, (4) approaches $[R - c(u)]_+$ uniformly as $a_r \downarrow 0$, with error at most $a_r k_j$ times the interval length. The node inverse is uniformly regular for sufficiently small $\beta \geq 0$ under (3). Equations (9)–(10) give continuous dependence on these kernels/parameters within valid regimes.

All prefix events used above have strict margins or transverse hits: E front shortfall, A birth in (20), fixed pulse end with positive queues, A clearance at positive service, and renewal fill at positive g_E. Between events local continuous dependence applies. At each hit a nonzero guard derivative bounds the hitting-time change; the compatible new regime then continues it. Induction through this finite event list preserves the strict inequalities for sufficiently small *positive* a_r and *positive* β. Oldest-profile loss at first retirement also stays positive. This removes both auxiliary zero parameters, and preserves an interval with $q_A > 0$ and nonzero continuing waiting area.

O-S started from the same inputs agrees through first fill t_R. At that shared state, physical aging and injection agree, but only O-R retires old weight. Since both budgets are positive,

$$(\dot{B}_{O\text{-}R} - \dot{B}_{O\text{-}S})(t_R+) = \frac{g_E(t_R)}{m} D(t_R; u_E(0+)) > 0. \tag{23a}$$

This is a consequential local difference, not a global trajectory or policy ordering. It does not establish that either closure is empirically correct.

After the finite pulse there are no new arrivals. Nonnegative queues are nonincreasing; at most two clearances remain. Bounds (11)–(13) make them finite; the clearance directions are compatible. The chosen small A tail makes A clear first with E still positive. E alone then clears at positive service. Once both queues are zero, no birth is allowed without arrivals; the fixed retained profiles simply age to zero. This proves a complete unique regular trajectory for the constructed family, not every possible switch sequence in the full parameter/action domain.

Realizing the inputs with unchanged H1. Choose $r = s/v_0$, $e_0 + a_0 = r$, $a_0 = b_0 > 0$. A receiving-limited initial branch followed by a strictly donor-limited front is possible when

$$k_c(v_0) - r < (\gamma - 1)e_0, \qquad e_f/v_0 < \{k_c(v_0) - r\}/(\gamma - 1). \tag{24}$$

Choose the front progress z_f after that switch. Increasing h while holding z_f/h fixed retains the same front density and makes $k = v_0/h$ small. Put $\ell = v_0 T$, $L_p = \ell + z_f$, and choose finite terminal length $b > M/k_j$, $M = \ell(r + b_0)$. This changes the common eligibility delay, not the translated node argument. Initial densities, $v_+ r < R$, $R \leq C(v_-)$ and an adequate independent B lane can all be retained.

The finite strict event sequence persists in a sufficiently small one-sided neighborhood of $v_0 = v_+$, inside the declared action interval. This is local forward coverage, not permission to shrink that interval or assume the remaining advices are qualified. Long geometry and very slow recovery may be needed by this proof. No applied parameter range, preferred advice or interior optimum is established.

8. Exact feasibility, accounting and the review checkpoint

An exact rational point verifies that the static inequalities are nonempty:

$$R = m = 1, \quad c_0 = 11/10, \quad e_f = 1/8, \quad \varepsilon = 1/512, \quad s = 511/512. \tag{25}$$

Then

$$\begin{aligned} d_0 = 2491/2816, \quad \mu = 12455/247808, \quad \varepsilon/\mu = 484/12455, \\ \sigma = 27/2816, \qquad (\varepsilon + e_f)/d_0 = 715/4982 < 1. \end{aligned} \tag{26}$$

Take $v_- = 1, v_0 = v_+ = 2, w = 2, k_j = 3$, $a_0 = b_0 = 1/8$, $e_0 = 383/1024$, $r = 511/1024$, $\gamma = 5$, $R_B = 1$. The fast initial receiving branch is strict; the slower one is donor-limited. A donor-limited fast front with density $1/16$ is obtained from

$$\frac{z_f}{h} = 5\log(5632/5125) + \log(1025/256). \tag{27}$$

k is subsequently chosen sufficiently small by the finite-prefix argument; no numerical traffic trajectory, terminal size or positive recovery/mismatch parameter set is selected. These values are algebraic feasibility evidence.

For the resulting admissible family, T1's storage proof is unchanged: $M < k_j b$ and $\dot{F}_j \le C(v)$ keep entrance vacancy positive. Early-merged mass enters A once. With remaining upstream occupied length s_{occ} and initial total density ρ_0,

$$s_{\mathrm{occ}}\rho_0 + \sum_j N_j + \sum_j Y_j = M, \qquad J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E)dt. \tag{28}$$

Retiring recovery information is not removing traffic. Post-line influence is not post-line vehicle-time. Known empty memory and initial continuing symmetry retain conditional total-plus-ending-statistic sufficiency for uniquely qualified joined cases. Neither full-lane-state necessity nor a positive information/action/gradient result follows.

Source boundary. The isolated point motivation is reused from Leclercq et al. (2016), Section 2; O-R and all proofs here are our adaptation and deductions. Jin's LTM boundary/node context does not make this a general invariant junction. The representative speed, renewal mass, maneuver/recovery partition and common-advice response still require physical qualification.

Checkpoint OR-Q1. The bounded J3 candidate-and-analytical-check pass is complete. O-R, R-E and S-E-P remain selected. The regular cohort advances mathematical coexistence, not applied MP6. Stop for review before J4's original-action-range, provenance and information/sensitivity/native qualification. Step 3 remains open, Step 4 unopened. No simulator, installation, traffic experiment, numerical gradient or optimization ran.